

List Comprehension in Python

- List comprehension is a concise way to create new lists by applying an expression to each item in an existing iterable like a list, tuple or range.
- It helps to write clean, readable and efficient code compared to traditional loops.

Suppose you want to square every number in a list:

```
a = [2, 3, 4, 5]
res = [val ** 2 for val in a]
print(res)
```

Output

```
[4, 9, 16, 25]
```

List Comprehension in Python

▪Syntax

[***expression*** for item in iterable]

```
a = [2, 3, 4, 5]  
res = [val*10 for val in a]  
print(res)
```

Output

[20, 30, 40, 50]

Conditional Statements in List Comprehens

▪Syntax

[***expression*** for item in iterable if condition]

Example 1 : This code uses a list comprehension with a condition to create a new list with only even numbers from list a.

```
a = [1, 2, 3, 4, 5]
res = [val for val in a if val % 2 == 0]
print(res)
```

Output

[2, 4]

Conditional Statements in List Comprehens

Example 2 : create a new list with numbers greater than 10 from list b.

```
b = [5, 12, 7, 18, 3, 20]  
res = [val for val in b if val > 10]  
print(res)
```

Output

[12, 18, 20]

List Comprehension

Example 3: Creating a list from a range

```
a = [i for i in range(10)]  
print(a)
```

Output

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
```

List Comprehension

Example 3: Two loops inside a list comprehension.

```
c = [(x, y) for x in range(3) for y in range(3)]  
print(c)
```

Output

```
[(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 1), (2, 2)]
```

List Comprehension

Create a list of tuple of a number and its square ?